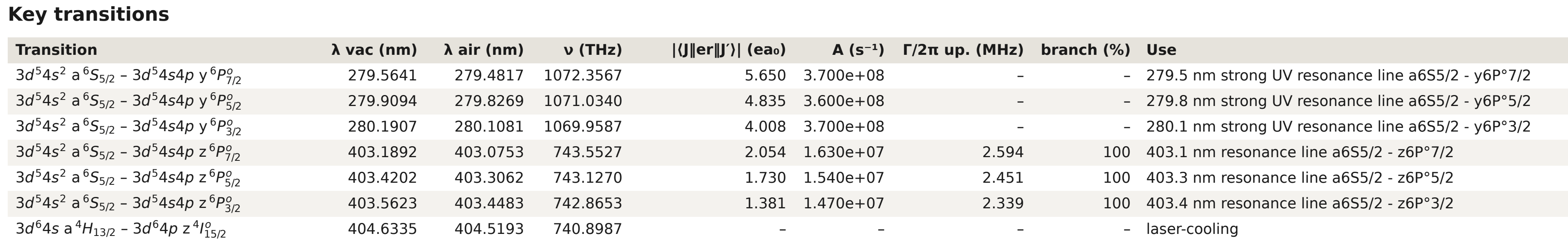


Mn⁺ ionisation limit 59959.56 cm⁻¹ = 7.43404 eV (166.779 nm)

Arrow colour = vacuum wavelength (nm)

300 400 500 600 700 800 1000 1200 1500 2000



Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)				
3d ⁴ 4s ² a ⁵ S ₂	-71.95	-	F=0: +629.56 F1= +557.61 F2= +413.71 F3= +197.86 F4= -89.94 F5= -449.69				
3d ⁴ 4s a ⁵ D _{3/2}	509.65	131.91	F2= -6947.23 F3= -5457.85 F4= -3456.63 F5= -930.36 F6= +2137.43 F7= +5766.54				
3d ⁴ 4s a ⁵ D _{5/2}	458.68	20.985	F1= -5148.91 F2= -4237.54 F3= -2868.25 F4= -1038.33 F5= +1255.82 F6= +4018.70				
3d ⁴ 4s a ⁵ D _{3/2}	437.70	-44.969	F0= -3861.35 F1= -3412.86 F2= -2519.92 F3= -1190.63 F4= +562.86 F5= +2724.38				
3d ⁴ 4s a ⁵ D _{5/2}	470.67	-65.954	F1= -2517.18 F2= -1523.08 F3= -81.39 F4= +1748.52				
3d ⁴ 4s a ⁵ D _{3/2}	881.390	-	F2= -1542.43 F3= +1101.74				
3d ⁴ 4s4p 2 ³ P _{3/2} ^o	572.60	11.992	F1= -2997.76 F2= -1862.15 F3= -149.75 F4= +2150.25				
3d ⁴ 4s4p 2 ³ P _{1/2} ^o	467.780	-74.948	F0= -4144.66 F1= -3659.00 F2= -2694.41 F3= -1264.38 F4= +610.83 F5= +2904.26				
3d ⁴ 4s4p 2 ³ P _{1/2} ^o	428.70	62.956	F1= -4789.15 F2= -3949.74 F3= -2683.87 F4= -983.46 F5= +1162.29 F6= +3766.86				
3d ⁴ 4s4p 2 ³ P _{3/2} ^o	971.330	17.988	F1= +5112.07 F2= +3155.02 F3= +232.94 F4= -3673.99				
3d ⁴ 4s4p 1 ³ P ₂ ^o	-539.630	-68.952	F0= +4673.50 F1= +4150.41 F2= +3408.05 F3= +1503.98 F4= -650.40 F5= -3389.93				
3d ⁴ 4s4p 1 ³ P ₂ ^o	-389.730	-185.870	F1= +4284.89 F2= +3558.53 F3= +2449.09 F4= +4932.65 F5= +1022.63 F6= -3456.61				
3d ⁴ 4p 2 ³ D _{3/2} ^o	86.940	-	F2= -1195.42 F3= -934.61 F4= -586.85 F5= -152.15 F6= +369.50 F7= +978.08				
3d ⁴ 4p 2 ³ D _{5/2} ^o	38.973	-	F1= -438.45 F2= -360.50 F3= -243.58 F4= -87.69 F5= +107.18 F6= +341.01				
3d ⁴ 4p 2 ³ D _{5/2} ^o	-26.981	-	F0= +236.08 F1= +209.10 F2= +155.14 F3= +74.20 F4= -33.73 F5= -168.63				
3d ⁴ 4p 2 ³ D _{3/2} ^o	-149.900	-	F1= +786.98 F2= +487.18 F3= +37.48 F4= -562.12				
3d ⁴ 4p 2 ³ D _{1/2} ^o	-821.430	-	F2= +1437.50 F3= -1026.79				
3d ⁴ 4p 2 ³ F _{7/2} ^o	179.880	-	F3= -2923.05 F4= -2203.53 F5= -1304.13 F6= -224.85 F7= +1034.31 F8= +2473.35				
3d ⁴ 4p 2 ³ F _{5/2} ^o	155.890	-	F2= -2143.49 F3= -1675.82 F4= -1052.26 F5= -272.81 F6= +662.53 F7= +1206.62				
3d ⁴ 4p 2 ³ F _{7/2} ^o	137.900	-	F1= -1551.38 F2= -1275.58 F3= -861.88 F4= -310.27 F5= +379.23 F6= +1256.62				
3d ⁴ 4p 2 ³ F _{5/2} ^o	173.880	-	F0= -1521.45 F1= -1347.57 F2= -999.81 F3= -478.17 F4= +217.35 F5= +1086.75				
3d ⁴ 4p 2 ³ F _{3/2} ^o	272.810	-	F1= -1432.25 F2= -886.63 F3= -68.20 F4= +1023.04				
3d ⁴ 4p 2 ³ F _{3/2} ^o	131.910	-	F2= -1813.76 F3= -1418.03 F4= -890.39 F5= -230.84 F6= +560.62 F7= +1483.99				
3d ⁴ 4p 2 ³ F _{1/2} ^o	170.880	-	F1= -1922.40 F2= -1580.64 F3= -1068.00 F4= -384.48 F5= +469.92 F6= +1495.20				
3d ⁴ 4p 2 ³ F _{1/2} ^o	281.800	-	F1= -3170.25 F2= -2606.65 F3= -1761.25 F4= -634.05 F5= +774.95 F6= +2465.75				
3d ⁴ 4p 2 ³ F _{3/2} ^o	290.800	-	F0= -2544.50 F1= -2253.70 F2= -1672.10 F3= -799.70 F4= +363.50 F5= +1817.50				
3d ⁴ 4p 2 ³ F _{3/2} ^o	374.740	-	F1= -1967.38 F2= -1217.90 F3= -93.69 F4= +1405.28				

Measured A, B constants; sources in the data file